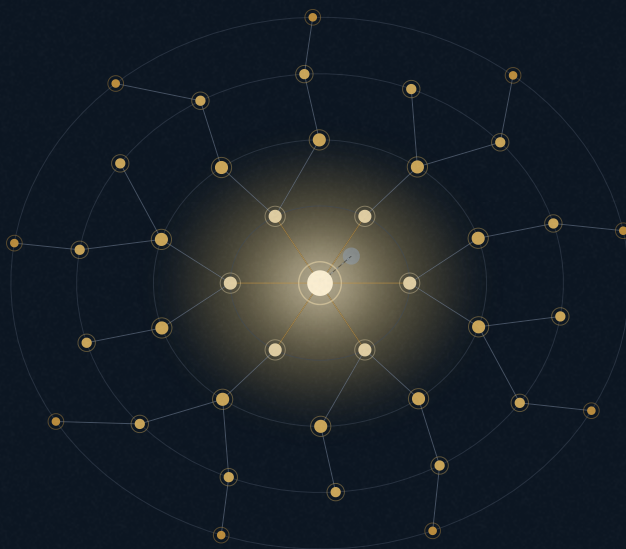




KRONOS FUSION ENERGY



PAPER 1.5 OF THE 2026 KRONOS SET

# *AI, ML & Quantum Control*

## *Certifiable Real-Time Control*

Physics-informed AI and quantum technologies for certifiable real-time control  
of compact fusion generators — three tiers, held apart.

Layered twin · 0/5000 escapes clamped · quantum: no crossover this decade

Explore it live — [3-D model](#) · [physics validator](#) · [film series](#)



THE OFFICIAL RECORD

# What this document is, and how to read its numbers

This is a combined editorial. It gathers, in one volume, the two design studies Kronos Fusion Energy released in 2026 — a compact spherical-tokamak tritium/helium-3 breeder and a deuterium–helium-3 tandem-mirror generator — together with the intellectual-property estate, the people, and, in the internal edition, the full commercial case. The scientific text is the same text that appears in the arXiv preprints and the journal submissions; the editorial adds the apparatus a reader needs to hold the whole program at once, and nothing in the physics is altered to fit it.

THE HONESTY STANCE IS THE ASSET

Every headline quantity carries an **evidence class** — DERIVED RECOMPUTED MEASURED REQUIREMENT — and, where a number is a modelled extrapolation rather than a demonstrated value, it is labelled as such and its downside is shown next to it. Several quantities in the underlying corpus moved after first publication; where a value has been withdrawn or restated, the record says so plainly. This is deliberate: a diligence team will find the load-bearing assumptions, so the document surfaces them first.

|                 |                                                                                                                                                                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TITLE           | The Kronos Fleet — Hyperion · Aegis · MetroVolt: A Combined Design & Commercialization Study                                                                                                                                                                       |
| AUTHOR          | Priyanca Ford, Founder & CEO, on behalf of Kronos Fusion Energy                                                                                                                                                                                                    |
| EDITION         | 2026 Combined Editorial · first issue                                                                                                                                                                                                                              |
| COMPANION WORKS | The five-paper arXiv drop — (1) Breeder, (2) Burner, (3) REBCO magnet & tape, (4) Direct Energy Conversion, (5) AI/ML/Quantum control — with matching numbered Zenodo deposits, plus the IOP journal and IAEA-FEC submissions. Patents were filed before the drop. |
| NAMING          | Hyperion = the breeder; Aegis (defense) and MetroVolt (data-center) = one generator in two housings. Internal mode letters (D1, L) do not appear on public surfaces.                                                                                               |
| STATUS          | Conceptual design & simulation study. Nothing herein is a construction commitment.                                                                                                                                                                                 |



HOW TO READ THIS VOLUME

# The fleet at a glance, and the way its numbers are marked

This volume opens with *Orientation*, presents the five research papers as a bound sequence, and closes with the *Apparatus*. *Foundations* sets out the three general results both machines rest on. *Paper One* is the Hyperion breeder; *Paper Two* the Aegis / MetroVolt generator; *Papers Three, Four and Five* are the enabling science — high-field REBCO magnets and tape, direct energy conversion, and the AI / ML / quantum control stack. A *Digital Twin & 3-D Model* part follows, then the *Environmental* profile. The *Economics* and levelized cost and the *Business Case* and diligence record appear in the internal edition only; the *Simulations* proof record and the Apparatus — patent portfolio, team, limitations, and sources — close both editions.

THE FLEET AT A GLANCE

|              | HYPERION                             | AEGIS                        | METROVOLT                       |
|--------------|--------------------------------------|------------------------------|---------------------------------|
| Role         | Strategic-isotope foundry            | Defense installation power   | Campus power                    |
| Machine      | Spherical tokamak                    | Tandem mirror                | Tandem mirror                   |
| Fuel         | Deuterium–tritium                    | Deuterium–helium-3           | Deuterium–helium-3              |
| Delivers     | Tritium · <sup>3</sup> He · 14 MeV n | Resilient installation power | Firm campus power               |
| Physics bar  | Fusion gain only                     | Closure (gates named)        | Closure + lunar <sup>3</sup> He |
| In the fleet | Breeds the fuel                      | Proves the generator         | Commercial destination          |

HOW THE NUMBERS ARE MARKED

Every headline quantity carries an evidence class — **DERIVED** from first principles, **RECOMPUTED** from raw data, **MEASURED** in hardware, or a **REQUIREMENT** yet to be met. Financial figures carry a case label and are confined to the internal (confidential) edition. Values that moved after first publication are shown as withdrawn or restated rather than quietly changed. A capability you can evaluate is one whose limits are on the page.

## ABSTRACT

# AI, ML & Quantum Control — Certifiable Real-Time Control

Real-time control of a nuclear-regulated fusion generator is treated as a layered digital-twin architecture in which physics-informed machine learning is deployable now — but only inside a deterministically clamped safe set. A control-barrier filter permits 0 safety-set escapes across 5000 trajectories against 5000 unfiltered, backed by an analytic proof. Quantum sensing is a near-term diagnostic tier; quantum *computation* is an offline calibration tier with no resource crossover this decade. Every miss is reported, including the equilibrium network's 1.4% against a 1% target.

Companion deposit (code & data, reproducible under CC BY 4.0): DOI [10.5281/zenodo.22132484](https://doi.org/10.5281/zenodo.22132484) · Zenodo community *kronos\_fusion\_energy*.

Explore it live: [interactive 3-D model](#) · [run the physics validator](#) · [companion film series](#) (the deposited solvers, in the browser).

**Keywords:**

digital twin

physics-informed ML

certifiable control

quantum sensing

safe-set clamp

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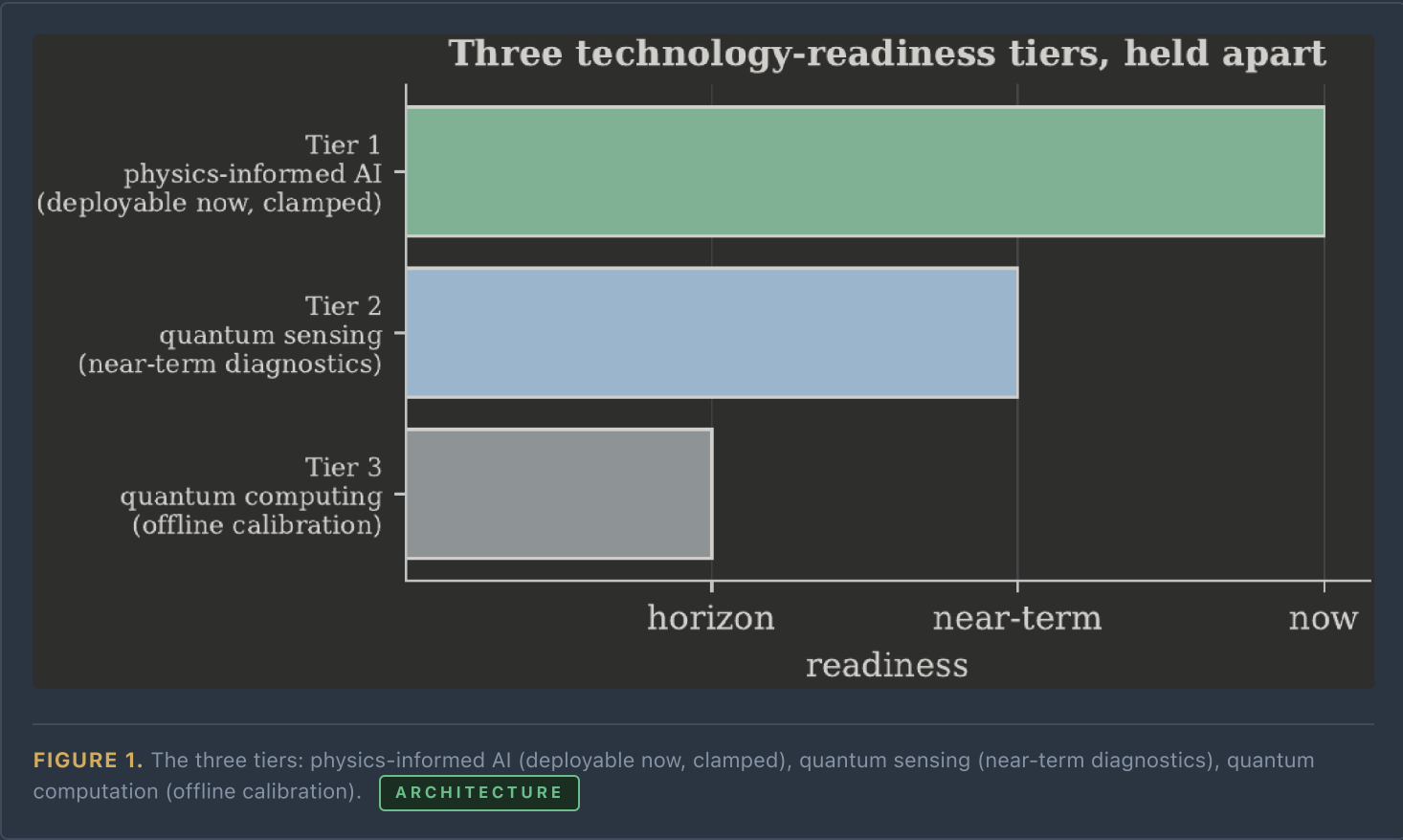
NOMENCLATURE

Symbols, units, and the names of things

|                  |                                                                        |
|------------------|------------------------------------------------------------------------|
| $Q$              | plasma (fusion) gain, $P_{\text{fus}}/P_{\text{aux}}$                  |
| $Q_E$            | engineering gain, net electric out / recirculating in                  |
| $H_{98}$         | confinement enhancement over the IPB98(y,2) scaling                    |
| $Z_{\text{eff}}$ | effective ion charge                                                   |
| $c_{ee}$         | electron–electron bremsstrahlung leading coefficient, 2.120022 (exact) |
| $\tau_E$         | energy confinement time                                                |
| $I_p$            | plasma current                                                         |
| $R_0, a$         | major radius, minor radius                                             |
| $\kappa, \delta$ | elongation, triangularity                                              |
| $\beta_N$        | normalized plasma beta (Troyon-normalized)                             |
| $TBR$            | tritium breeding ratio                                                 |
| $DEC$            | direct energy conversion                                               |
| $CF$             | plant capacity factor (availability)                                   |
| $FOAK/NOAK/BOAK$ | first / n <sup>th</sup> / best of a kind                               |
| <i>Hyperion</i>  | the ST breeder (internal: Mode D2)                                     |
| <i>Aegis</i>     | the generator, defense/installation housing (internal: Mode M)         |
| <i>MetroVolt</i> | the generator, data-center housing (Mode M)                            |

# Three tiers, held apart

A nuclear-regulated fusion generator must be controlled across ten orders of magnitude in time — microsecond magnet protection through month-scale fuel logistics — and every automated actuation must be bounded, auditable, and reproducible. Machine learning and quantum technologies are often invoked for this without saying what is deployable now versus what is a decade away. We hold three technology-readiness tiers strictly apart, and report each with its evidence *and* its misses.



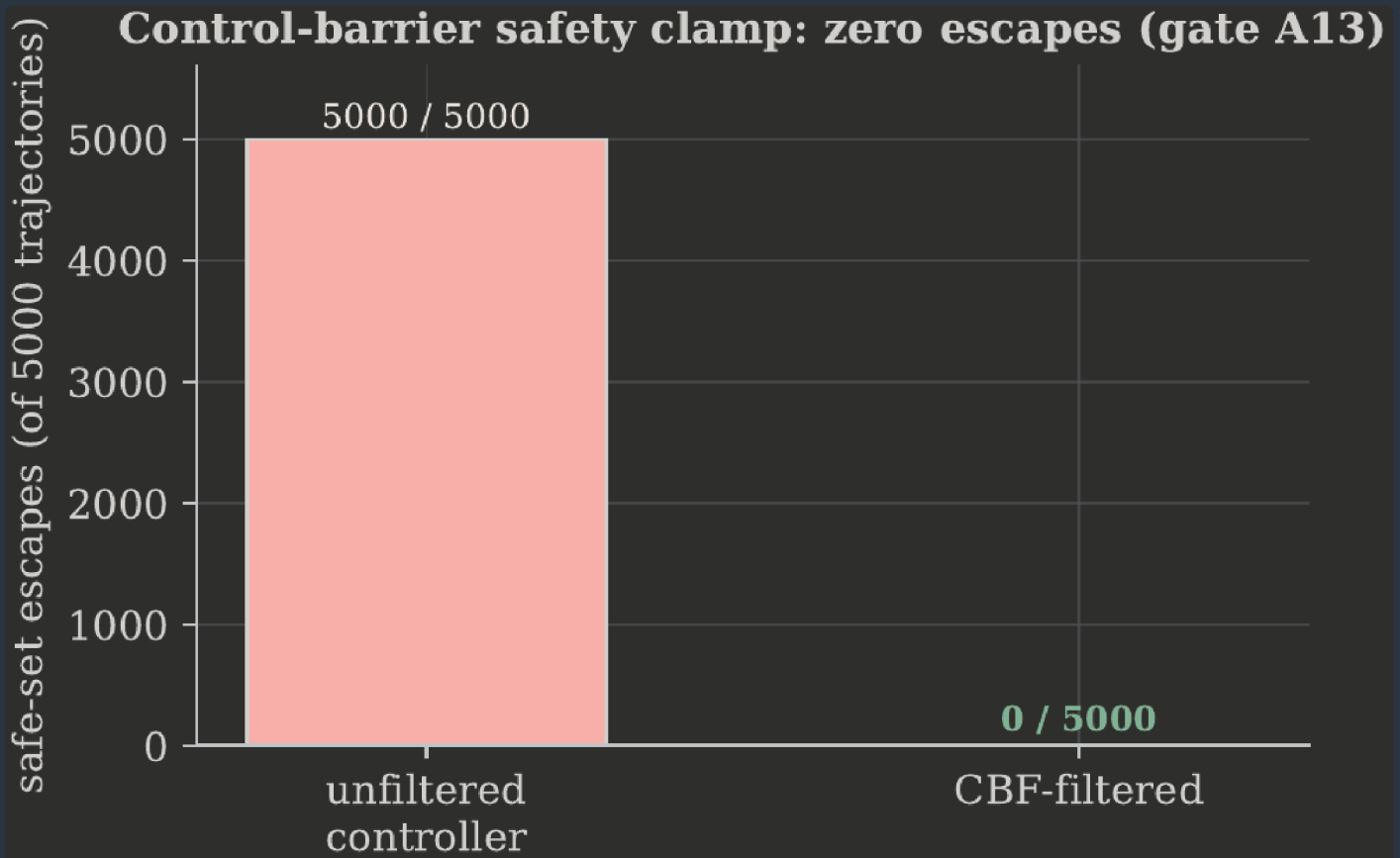
## § 2

## Tier 1 — physics-informed AI, deployable now if clamped

The load-bearing requirement is that a learned controller cannot drive the machine out of its safe operating envelope. A robust control-barrier-function filter provides this with an analytic forward-invariance guarantee: over 5000 Monte-Carlo trajectories, 0 escape the safe set with the filter against 5000 of 5000 without it. Around it, a Bayesian-optimization surrogate reaches leave-one-out  $R^2 = 0.987$ . Physics-informed AI is deployable in the loop *because* its error is bounded and deterministically clamped.

### WHY THIS IS THE LOAD-BEARING RESULT

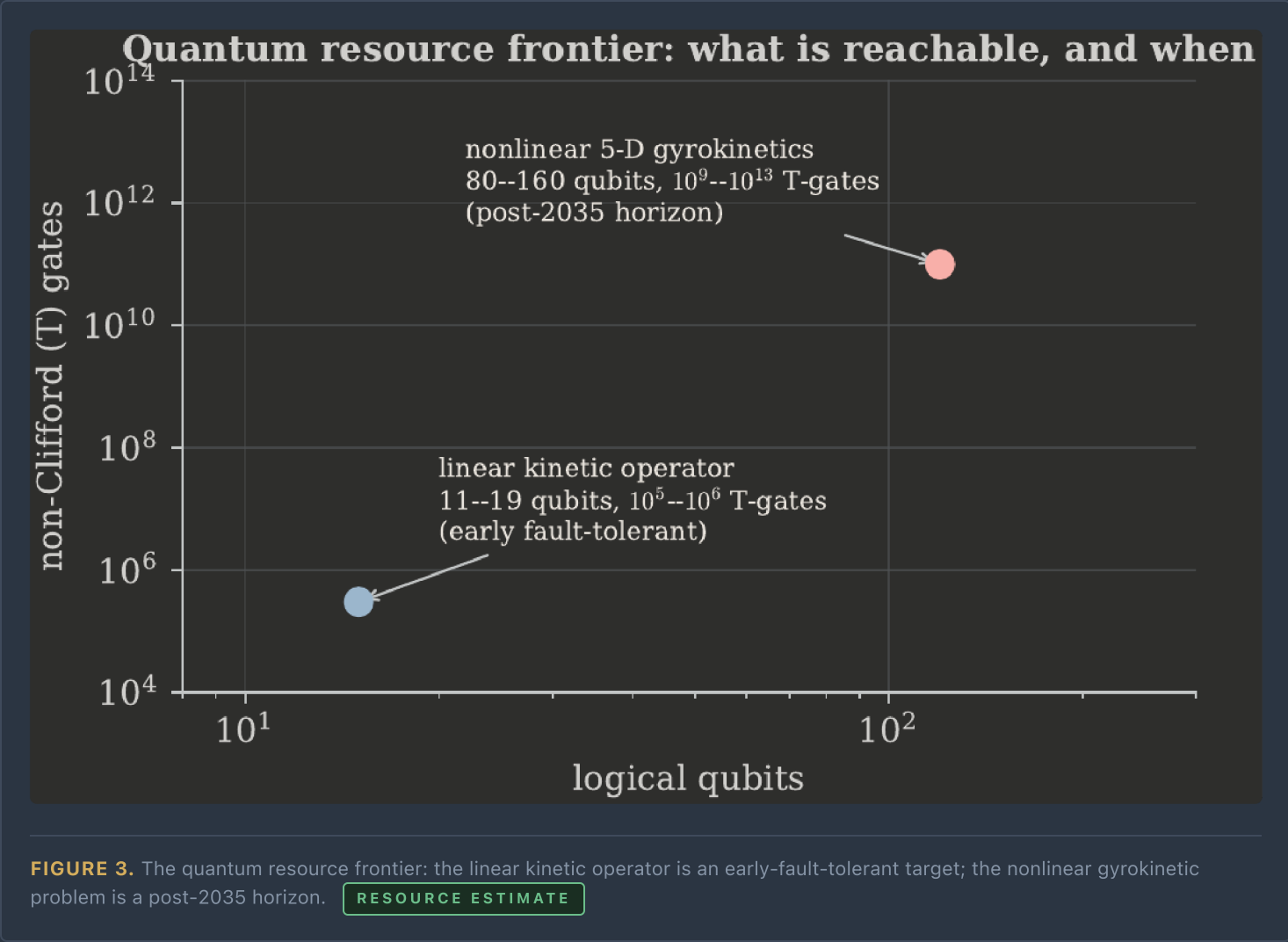
A learned controller can pursue performance while a proven clamp guarantees the machine stays in the safe envelope — zero safety-set escapes with the filter, backed by proof, not by a training score.



**FIGURE 2.** The control-barrier safety clamp: 0 of 5000 trajectories escape the safe set with the filter, against 5000 unfiltered. **MONTE-CARLO**

## Tier 3 — quantum computation, an offline calibration tier

For fusion, quantum computation is an offline calibration tier with no resource crossover this decade. Landau damping reproduces on a 5-qubit circuit to a few percent, and variational quantum chemistry reaches chemical accuracy on the fusion-alloy fragments — but the honest resource split is load-bearing: the linear kinetic operator needs 11–19 logical qubits, and the nonlinear gyrokinetic problem that would most help the machine is fault-tolerant-horizon. Noisy hardware today shows no advantage; the classical baseline is cheaper at every demonstrated scale.



# Reporting the misses

We report every missed pre-registered target rather than reframing it. The physics-informed equilibrium network was intended to become a fast native solver and did not: it reached **1.4%** relative- $L^2$  error against a 1% goal. Keeping this negative in the record is part of the method. The contribution is a certifiable, resource-honest system and method for compact-fusion control — not new plasma physics.

REPORTED, NOT REFRAMED

APPARATUS

## *Portfolio, People & Record*

*The intellectual-property estate, the people who built it, the limitations that gate it, and the sources it rests on.*

1

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INTELLECTUAL PROPERTY

The patent portfolio

The estate spans a granted high-field magnet patent, pending utility and 2026 provisional filings that map to the two products, a digital-twin control provisional, a registered trademark application, and the original 2022 provisional family. Every patentable disclosure in the five-paper arXiv drop was protected **before** publication: the breeder and burner provisionals were filed 1–2 August, and a publication-gap omnibus filing the evening before the drop swept up the DEC, plasma-control, and REBCO-winding matter of the three otherwise-unprotected papers. Patent numbers and application serials are matters of public record; claim scope is summarised, not reproduced.

GRANTED & PENDING UTILITY

|               |                                                                                   |         |
|---------------|-----------------------------------------------------------------------------------|---------|
| US 12,009,112 | High-field tilted graded-REBCO magnet architecture (KRONO-002A) · app. 17/878,550 | GRANTED |
| US 17/878,507 | Core device / method (KRONO-001A) · utility                                       | PENDING |

2026 PROVISIONAL FILINGS — MAPPED TO THE PRODUCTS

|            |                                                                                                                                                                                                                                |       |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 64/124,209 | Hyperion — spherical-tokamak tritium / helium-3 foundry · breeder · filed Aug 2026                                                                                                                                             | FILED |
| 64/124,220 | Aegis / MetroVolt — synchrotron-cutoff, fuel-selection & plug-winding · generator · filed Aug 2026                                                                                                                             | FILED |
| 64/115,167 | KRONOS-CTRL digital-twin plant control · provisional · filed Jul 2026                                                                                                                                                          | FILED |
| 64/005,440 | Integrated spherical-tokamak fusion architecture · early integrated-architecture filing · filed Mar 2026                                                                                                                       | FILED |
| 64/105,530 | Negative-triangularity spherical-tokamak architecture · foundational ST filing · filed Jul 2026                                                                                                                                | FILED |
| 64/128,097 | Publication-gap omnibus — quasineutral two-species expander DEC · provenance-gated / latency-split plasma control · as-built high-field winding & conductor architecture · filed 7 Aug 2026, the evening before the arXiv drop | FILED |

TRADEMARK & ORIGIN

|              |                                                                          |          |
|--------------|--------------------------------------------------------------------------|----------|
| FTK-98801894 | Kronos Fusion Energy — trademark application                             | FILED    |
| KR-2022-★    | Original provisional family (KR-2022-00001 ... 00027) · 27 filings, 2022 | PRIORITY |

The narrow, defensible magnet novelty — the specific conductor and the digital-twin winding optimisation, not high-field REBCO as a category — is the commercial core that can earn ahead of any  $Q > 1$  milestone, with markets in fusion magnet supply, MRI/NMR, accelerators, and proton therapy. The claim is scoped honestly: the high field is a *system field* result, not a stand-alone-coil record; the small-bore plug coil is structurally infeasible as a bare winding but resolved by a stress-managed structural shield (feasible-pending-FEA); and the winding-tape experiment returned a *null result*. The value is the method, not a field record.

## THE TEAM

## Founding partners, board, advisors &amp; auditors

Kronos is built by a bench of advanced-fuel-fusion, high-field-magnet, direct-conversion, and materials specialists, with a board and operations team drawn from defense, national laboratories, and industry. Dates are shown as ranges; where a tenure has ended or is term-ending, the range says so.

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Legal counsel is retained; those roles are held on the internal roster and are not listed here.

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## COLOPHON

# How this document was made

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*The Kronos Fleet — Combined Editorial, 2026* is set in **Fraunces** (display), **Gelasio** (text), and **IBM Plex Mono** (data & equations), carried forward from the Kronos editorial design system.

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The figures are rendered at 300 dpi from the deposited generator scripts; the document is composed as a single self-contained file with embedded fonts and imagery, paginated to US Letter, and rendered to PDF through a headless Chromium engine in matched light and dark editions.

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Every headline quantity carries an evidence class; withdrawn and restated values are marked as such. Nothing herein is frozen beyond the founder-approved Tier-A set.

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**Numbers and their classes.** Quantities are reported with evidence classes and case labels; modelled, assumed, and requirement-class values are identified as such and are not to be quoted without their labels.

**Public edition.** This edition carries the physics and design only; all financial, funding, and defense-commercial content has been removed for public distribution. The scientific text corresponds to the arXiv and journal submissions.

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THE 2026 KRONOS SET

# One programme, in five papers

This editorial is one of five that together describe the Kronos fusion programme — a breeder that funds a generator, and the magnet, conversion, and control science that enable both. Each is published open-access with a reproducible companion deposit.

|           |                                                                                                                           |
|-----------|---------------------------------------------------------------------------------------------------------------------------|
| PAPER I   | Hyperion — The Tritium Foundry · <a href="https://zenodo.org/record/21746157">10.5281/zenodo.21746157</a>                 |
| PAPER II  | Aegis & MetroVolt — The Generator · <a href="https://zenodo.org/record/21746479">10.5281/zenodo.21746479</a>              |
| PAPER III | High-Field REBCO Magnets & Tape · <a href="https://zenodo.org/record/21842514">10.5281/zenodo.21842514</a>                |
| PAPER IV  | Direct Energy Conversion · <a href="https://zenodo.org/record/21842864">10.5281/zenodo.21842864</a>                       |
| PAPER V   | AI, ML & Quantum Control ( <i>this paper</i> ) · <a href="https://zenodo.org/record/21842371">10.5281/zenodo.21842371</a> |

Explore the whole programme live: the interactive [3-D model](#), the in-browser [physics validator](#), and the [companion film series](#).



KRONOS FUSION ENERGY

# *AI, ML & Quantum Control*

Explore the science — live, in your browser.

[kronosfusionenergy.com/3D-Model](https://kronosfusionenergy.com/3D-Model)

[kronosfusionenergy.com/physics-Validation-Simulation](https://kronosfusionenergy.com/physics-Validation-Simulation)

[Companion film series · YouTube](#)

[Zenodo · DOI 10.5281/zenodo.22132484](https://doi.org/10.5281/zenodo.22132484)